

Engineering Physics for Applied Electrical Technology and Computing

Interactive Theory, Laboratory, Numerical and Examination Handbook with Malayalam concept support.

Malayalam: Mechanics, Fluid Dynamics, Rotational Motion, Wave Motion, Semiconductor Physics, Photoelectric Effect, Laser എന്നീ concepts theory, numerical problems, diagrams, animations, laboratory experiments എന്നിവയുമായി ബന്ധപ്പെട്ട് ലളിതമായി പഠിക്കാനായുള്ള വലിയ handbook ആണ്.

Semester shown in source	I
Teaching Scheme	3:0:3:0
Contact Hours	90
Credits	4.5
CIE	40
ESE	60
Modules	4
Minimum Experiments	8-10

COURSE DASHBOARD

What this handbook teaches

Applied Physics connects measurable physical laws with electrical technology, computing hardware, instrumentation and engineering practice. The handbook follows the official Revision 2026 Course 2002B syllabus and combines theory, experiments, formulas, diagrams, numericals and examination preparation.

Malayalam overview: Physics laws മനപ്പാഠമാക്കുന്നത് മാത്രം അല്ല; ഓരോ concept-ഉം electrical technology, computing, measurement, laboratory work എന്നിവയിൽ എങ്ങനെ ഉപയോഗിക്കുന്നു എന്ന് diagrams, solved numericals, experiments എന്നിവയിലൂടെ മനസ്സിലാക്കുകയാണ് ഈ handbook-ന്റെ ലക്ഷ്യം.

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Apply mechanics. Newton's Laws, Momentum, Work, Energy and Power.

Apply Elasticity, Fluid Dynamics, Viscosity and Heat Transfer.

CO3

Analyse Circular Motion, Rotation, SHM and Waves.

CO4

Apply Semiconductor Physics, Photoelectric Effect and Lasers.

Source note: The supplied official course PDF displays Semester I. Programme catalogue placement must remain the website source of truth if it differs. The self-learning table also includes Optics, Electricity and Magnetism activities outside the listed Module III core theory; they are retained below as official enrichment.

Course objectives

- Develop broad understanding of physical principles and quantitative reasoning.
- Apply Physics to broad engineering problems.
- Gain first-hand experience with apparatus.
- Build confidence in measurement and equipment handling.

Assessment

- Theory CIE: 20
- Theory ESE: 60
- Practical CIE: 20
- Official total shown in detailed scheme: 100

Study method

1. Read concept and Malayalam help.
2. Study diagram or animation.
3. Work the solved example.
4. Complete practical record.
5. Answer expected questions.

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CO	PO1	PO2	PO3	PO6	PO7	PO8
CO1	3	2	2	2	2	2
CO2	3	2	2	2	2	2
CO3	3	2	2	2	2	2
CO4	3	2	2	2	2	2

LEARNING ROADMAP

From measurement to modern physics

FORMULA BANK

Core equations, symbols and use

Linear Motion

$$v = u + at \quad ns = ut + \frac{1}{2}at^2 \quad nv^2 = u^2 + 2as$$

Use only for uniform acceleration.

Force & Momentum

$$F = ma \quad np = mv \quad nm_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Momentum is conserved in an isolated system.

Work, Energy, Power

$$W = Fs \cos\theta \quad nKE = \frac{1}{2}mv^2 \quad nPE = mgh \quad nP = W/t$$

Elasticity

$$\text{Strain} = \Delta L/L \quad nY = \text{Stress/Strain}$$

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Fluid Dynamics

$$A_1 v_1 = A_2 v_2 \quad P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant} \quad F = 6\pi\eta r v$$

Temperature

$$F = (9/5)C + 32 \quad C = (5/9)(F - 32) \quad K = C + 273.15$$

Circular Motion

$$v = r\omega \quad a_t = r\alpha \quad a^c = v^2/r = r\omega^2 \quad F^c = mv^2/r$$

Rotation

$$I = \sum mr^2 \quad I = Mk^2 \quad \tau = rF \sin\theta \quad L = I\omega$$

Axis Theorems

$$I = I_{cm} + Md^2 \quad I_z = I_x + I_y$$

State the axis in every answer.

Waves & Oscillations

$$v = f\lambda \quad T = 1/f \quad T_{\text{pendulum}} = 2\pi\sqrt{l/g} \quad T_{\text{spring}} = 2\pi\sqrt{m/k}$$

Photon

$$E = hf = hc/\lambda$$

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Photoelectric Effect

$$hf = \phi + KE_{\max}$$

Emission occurs only when frequency exceeds threshold frequency.

MODULE I

Mechanics

10 Theory Hours

CO1

4 Listed Experiments

Malayalam overview: Units, Vector, Motion, Newton’s Laws, Momentum, Work, Energy, Power എന്നിവ engineering motion problems analyse ചെയ്യാൻ ഉപയോഗിക്കുന്ന രീതിയാണ് ഈ Module വിശദീകരിക്കുന്നത്.

1. Units, Fundamental and Derived Quantities –

A **physical quantity** is a measurable property expressed by a numerical value and unit. Fundamental quantities are independent; derived quantities are formed from them.

Fundamental quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Examples of derived quantities: velocity (m/s), acceleration (m/s²), force (N = kg·m/s²), work (J = N·m), power (W = J/s).

Malayalam: Physical Quantity എന്നത് അളക്കാൻ കഴിയുന്ന ഒരു property ആണ്. Fundamental Quantities മറ്റൊരു quantity ഉപയോഗിച്ച് നിർവചിക്കാത്ത അടിസ്ഥാന quantities ആണ്.

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2. Scalars, Vectors and Resolution —

A scalar has magnitude only; a vector has magnitude and direction. Vector addition follows Triangle Law and Parallelogram Law. A vector A at angle θ has rectangular components $A_x = A \cos\theta$ and $A_y = A \sin\theta$.

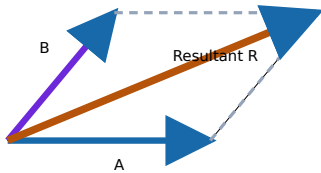
Scalars: mass, time, temperature, speed, energy.

Vectors: displacement, velocity, acceleration, force, momentum.

Triangle Law: place the tail of the second vector at the head of the first.

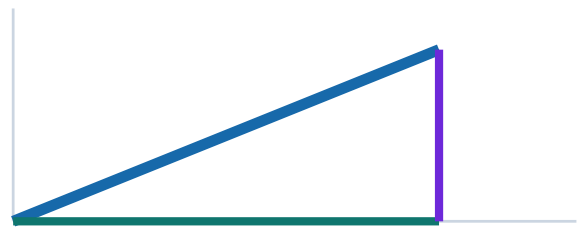
Parallelogram Law: diagonal through the common origin gives the resultant.

Malayalam: Vector-ന് magnitude-ഉം direction-ഉം ഉണ്ടായിരിക്കും. Resolution of Vector എന്നത് Vector-നെ horizontal, vertical components ആയി വേർതിരിക്കുന്നതാണ്.



Parallelogram Law: diagonal represents the resultant.

Vector component animation



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3. Displacement, Velocity, Acceleration and Equations of Motion—

Distance is total path length; displacement is the shortest directed change in position. Speed is distance/time, velocity is displacement/time, and acceleration is rate of change of velocity.

$$v = u + at \quad ns = ut + \frac{1}{2}at^2 \quad nv^2 = u^2 + 2as$$

Solved example

A vehicle starts from rest and accelerates at 2 m/s^2 for 5 s.

Given: $u=0$, $a=2$, $t=5$

v: $0+(2 \times 5)=10 \text{ m/s}$

s: $0+(\frac{1}{2} \times 2 \times 25)=25 \text{ m}$

Motion calculator

Initial velocity u (m/s)

0

Acceleration a (m/s^2)

2

Time t (s)

5

Enter values and calculate.

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4. Newton's Laws, Momentum and Recoil—

First Law: a body remains at rest or uniform straight-line motion unless acted upon by an unbalanced external force.

Second Law: force equals rate of change of momentum; for constant mass, $F=ma$.

Third Law: every action has an equal and opposite reaction acting on another body.

Momentum: $p=mv$, unit $\text{kg}\cdot\text{m/s}$.

For an isolated system, total momentum before interaction equals total momentum after interaction. In gun recoil, forward bullet momentum equals backward gun momentum in magnitude.

Malayalam: Inertia എന്നത് ഒരു body അതിന്റെ നിലവിലെ state മാറ്റത്തെ എതിർക്കുന്ന property ആണ്. Recoil-ൽ bullet-ന്റെ forward momentum-ന് തുല്യമായ opposite momentum gun-ന് ലഭിക്കുന്നു.

Recoil animation



Bullet momentum

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5. Work, Energy, Conservation of Energy and Power—

Work is done when a force causes displacement: $W = F_s \cos\theta$. Work is positive when force assists displacement, negative when it opposes it, and zero when force is perpendicular to displacement or displacement is zero.

Kinetic Energy is energy due to motion, $KE = \frac{1}{2}mv^2$. Gravitational Potential Energy is $PE = mgh$. In ideal free fall, PE decreases while KE increases, but total mechanical energy remains constant. Power is rate of doing work, $P = W/t$, measured in watt.

Malayalam: Freely falling body താഴേക്ക് വരുമ്പോൾ Potential Energy കുറയുകയും Kinetic Energy കൂടുകയും ചെയ്യും. Ideal condition-ൽ total Mechanical Energy constant ആണ്.

Free-fall energy transformation



Potential Energy

Kinetic Energy

MODULE II

Properties of Matter and Thermodynamics

10 Theory Hours

CO2

4 Listed Experiments

Malayalam overview: Elasticity, Stress, Strain, Fluid Flow, Bernoulli's Theorem, Viscosity, Heat Transfer എന്നിവ material-ുകളുടെയും fluid-ുകളുടെയും practical behaviour മനസ്സിലാക്കാൻ സഹായിക്കുന്നു.

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1. Elasticity, Stress, Strain and Hooke's Law—

Elasticity is the property by which a body regains original dimensions after removal of deforming force, within its elastic limit. Stress is internal restoring force per unit area; strain is fractional change in dimension.

Stress = Force/Area
Strain = Change in dimension/Original dimension
Young's Modulus = Longitudinal Stress/Longitudinal Strain
Rigidity Modulus = Shearing Stress/Shearing Strain
Bulk Modulus = Volume Stress/Volume Strain

Hooke's Law: within the elastic limit, stress is directly proportional to strain. For a spring, $F=kx$.

Malayalam: Stress എന്നത് unit area-ൽ പ്രവർത്തിക്കുന്ന internal restoring force ആണ്. Strain എന്നത് dimension-ൽ വന്ന change-ന്റെ original dimension-നോടുള്ള ratio ആണ്.

Solved example

A 2 m wire of area 1 mm^2 extends by 1 mm under 100 N.

$$Y = (F/A)/(\Delta L/L) = (100/1 \times 10^{-6})/(1 \times 10^{-3}/2) = 2 \times 10^{11} \text{ Pa.}$$

2. Streamline Flow, Turbulent Flow and Continuity—

In streamline flow, every fluid particle follows a smooth path and velocity at a point is steady. Turbulent flow has irregular eddies and rapidly changing velocity. For incompressible steady flow, mass conservation gives $A_1V_1=A_2V_2$.

Malayalam: Pipe-ന്റെ cross-sectional area കുറയുമ്പോൾ Fluid Velocity കൂടുന്നു. ഇത് Equation of Continuity കാണിക്കുന്നു.



Streamline flow through a constriction

Velocity increases where the flow area decreases.

Streamline / Turbulent comparison



Streamline



Turbulent

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3. Bernoulli's Theorem and Atomizer—

For steady flow of an ideal fluid along a streamline, pressure energy, kinetic energy and potential energy per unit volume remain constant.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

In an atomizer, fast air across the top of a tube produces lower pressure. Higher pressure on the liquid surface pushes liquid up the tube and breaks it into droplets.

Malayalam: Fluid Velocity കൂടുന്ന ഭാഗത്ത് pressure കുറയാം. ഈ pressure difference ഉപയോഗിച്ചാണ് Atomizer liquid മുകളിലേക്ക് ഉയർത്തി spray ചെയ്യുന്നത്.

Atomizer principle



4. Viscosity, Viscous Force and Stoke's Law—

Viscosity is internal resistance to relative motion between fluid layers. The coefficient of viscosity η measures this resistance. For a small sphere moving slowly through a viscous liquid:

$$F = 6\pi\eta rv$$

At terminal velocity, net force becomes zero and the sphere moves with constant speed. Viscosity of liquids generally decreases as temperature increases.

Malayalam: Viscosity ഒരു fluid-ന്റെ flow-നെ എതിർക്കുന്ന internal resistance ആണ്. Honey-ക്ക് water-നെക്കാൾ viscosity കൂടുതലാണ്.

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5. Heat, Temperature, Scales and Heat Transfer—

Heat is energy transferred due to temperature difference; temperature indicates degree of hotness and determines heat-flow direction. Common scales are Celsius, Fahrenheit and Kelvin.

$$F = (9/5)C + 32 \quad C = (5/9)(F - 32) \quad K = C + 273.15$$

Temperature converter

Celsius

Enter °C and convert.

Modes of heat transfer

- **Conduction:** transfer through particle interaction, mainly in solids.
- **Convection:** transfer by bulk motion of a fluid.
- **Radiation:** transfer by electromagnetic waves; no medium required.

Malayalam: Conduction-ൽ bulk movement ഇല്ല. Convection-ൽ fluid movement ഉണ്ട്. Radiation-ന് medium ആവശ്യമില്ല.

MODULE III

Rotational and Wave Motion

11 Theory Hours

CO3

5 Listed Experiments

Malayalam overview: Circular Motion, Moment of Inertia, Torque, Angular Momentum, SHM, Wave Motion എന്നിവ rotating machines, vibration, sound എന്നീ applications-നോട് ബന്ധപ്പെടുത്തി പഠിക്കുന്നു.

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1. Circular Motion and Angular Quantities—

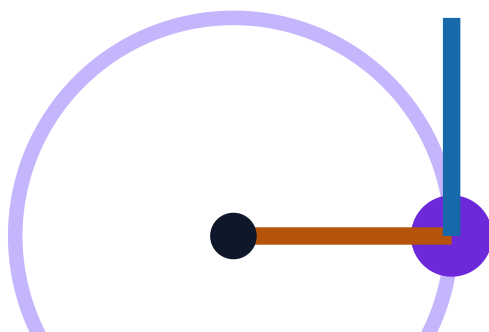
Circular motion is motion along a circular path. Angular displacement θ is measured in radian; angular velocity $\omega = d\theta/dt$; angular acceleration $\alpha = d\omega/dt$.

$$v = r\omega \quad a_t = r\alpha \quad a_c = v^2/r = r\omega^2 \quad F_c = mv^2/r = mr\omega^2$$

Velocity is tangential, while centripetal acceleration and force point toward the centre.

Malayalam: Circular Motion-ൽ velocity direction തുടർച്ചയായി മാറുന്നു. Centre-ലേക്ക് directed ആയ acceleration ആണ് Centripetal Acceleration.

Circular motion vectors



2. Rigid Body, Moment of Inertia and Radius of Gyration—

A rigid body is an ideal body whose particle separations remain constant. Moment of Inertia measures rotational inertia and depends on mass distribution and chosen axis.

$$I = \sum mr^2 \quad I = Mk^2$$

Malayalam: Moment of Inertia rotational motion-ലുള്ള inertia ആണ്. Mass axis-ൽ നിന്ന് കൂടുതൽ അകലെയായാൽ Moment of Inertia കൂടും.

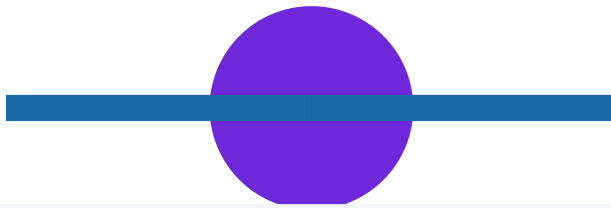
Standard Moment of Inertia formulae

Body	Axis	I
Thin rod	Through centre, perpendicular to length	$ML^2/12$
Thin rod	Through one end, perpendicular to length	$ML^2/3$
Disc	Central axis perpendicular to plane	$MR^2/2$
Ring	Central axis perpendicular to plane	MR^2
Skip to handbook	Diameter	$2MR^2/5$
Hollow sphere	Diameter	$2MR^2/3$

3. Axis Theorems, Torque and Angular Momentum—

Parallel Axis Theorem: $I = I_{cm} + Md^2$. **Perpendicular Axis Theorem:** for a plane lamina, $I_z = I_x + I_y$. Torque is turning effect, $\tau = rF \sin\theta$. Angular momentum is $L = I\omega$ and remains constant when external torque is zero.

Conservation of angular momentum



4. Periodic Motion and Simple Harmonic Motion—

Periodic motion repeats after equal time intervals. In SHM, restoring force is proportional to displacement and directed toward mean position. Important quantities are amplitude, time period and frequency.

For simple pendulum: $T = 2\pi\sqrt{l/g}$
For spring-mass system: $T = 2\pi\sqrt{m/k}$

Malayalam: SHM-ൽ Restoring Force displacement-ന് proportional ആയിരിക്കുകയും Mean Position-ലേക്ക് directed ആയിരിക്കുകയും ചെയ്യും.

Simple harmonic motion



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5. Transverse and Longitudinal Waves—

A wave transfers energy without net transport of matter. In transverse waves, particles vibrate perpendicular to propagation; in longitudinal waves, particles vibrate parallel, forming compressions and rarefactions.

$$v = f\lambda \quad T = 1/f$$

Wave characteristics: velocity, time period, frequency, wavelength, amplitude and phase.

Malayalam: Transverse Wave-ൽ particles wave direction-ന് perpendicular ആയി vibrate ചെയ്യും. Longitudinal Wave-ൽ particles wave direction-ന് parallel ആയി vibrate ചെയ്യും.

Wave calculator

Frequency f (Hz)

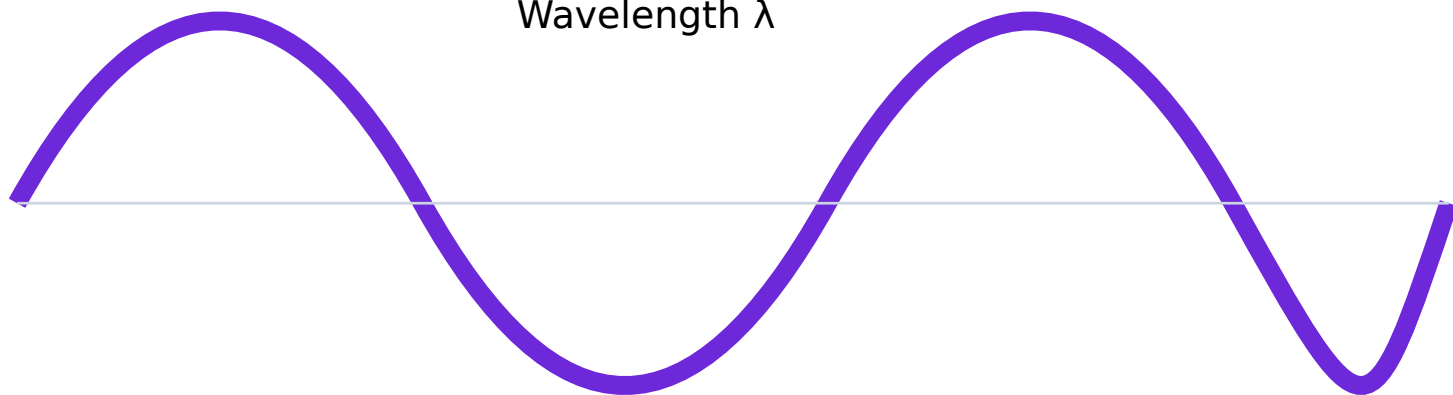
500

Wavelength λ (m)

0.68

Enter f and λ .

Wavelength λ



Transverse wave showing amplitude and wavelength.

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MODERN TV

Modern Physics

Malayalam overview: Semiconductor, Doping, PN Junction, Photoelectric Effect, Photon, Laser എന്നീ concepts electronic devices, sensors, computing hardware എന്നിവയുടെ അടിസ്ഥാനമാണ്.

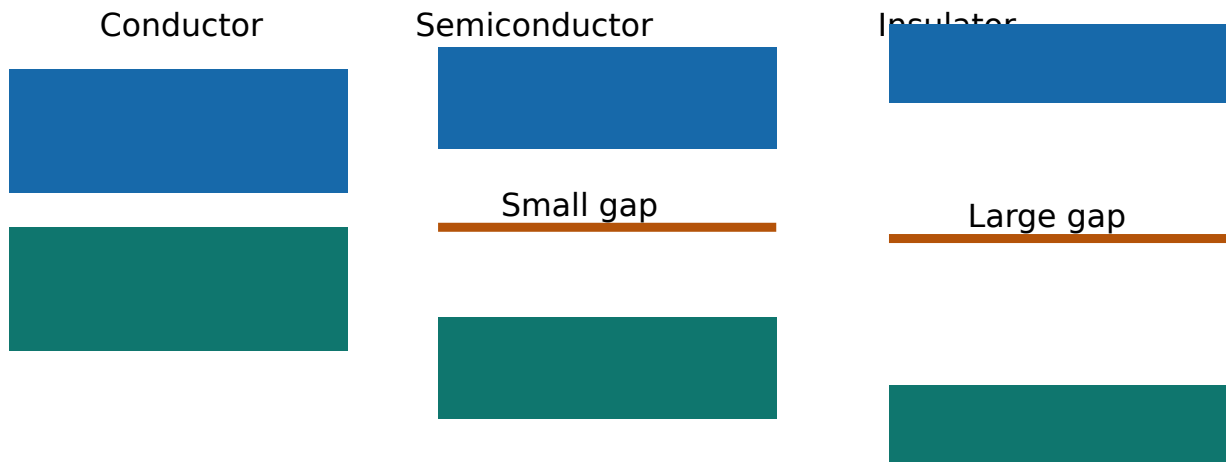
1. Energy Bands and Classification of Solids—

In solids, closely spaced atomic energy levels form bands. The highest occupied band is the Valence Band; the next allowed band is the Conduction Band. The forbidden gap determines electrical behaviour.

Conductor: bands overlap or conduction band is partly filled.

Semiconductor: small forbidden gap.

Insulator: large forbidden gap.



Energy-band comparison of conductor, semiconductor and insulator.

2. Intrinsic, Extrinsic, N-type and P-type Semiconductors—

An intrinsic semiconductor is pure silicon or germanium. Thermal energy creates electron-hole pairs. Doping adds controlled impurity atoms. Pentavalent donor impurity forms N-type material with electrons as majority carriers; trivalent acceptor impurity forms P-type material with holes as majority carriers.

Malayalam: Pure Semiconductor-ലേക്ക് ചെറിയ അളവിൽ impurity ചേർത്ത് conductivity മാറ്റുന്ന process ആണ് Doping. N-type-ൽ electrons majority carriers ആണ്; P-type-ൽ holes majority carriers ആണ്.

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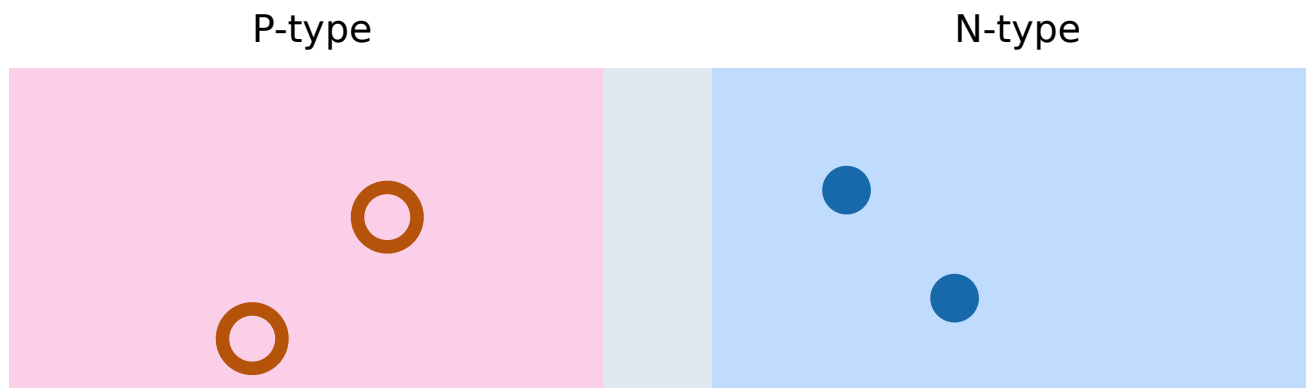
3. PN Junction, Forward Bias, Reverse Bias and V-I Characteristics —

When P-type and N-type materials join, carrier diffusion and recombination produce a depletion region and barrier potential. In forward bias, P is connected to positive and N to negative, reducing the barrier. In reverse bias, the barrier increases and only a small reverse saturation current flows.

The diode V-I curve contains forward knee voltage, rapid forward current rise and small reverse current. Typical knee voltage is about 0.7 V for silicon and 0.3 V for germanium.

Malayalam: Forward Bias barrier കുറച്ച് current flow വർദ്ധിപ്പിക്കുന്നു. Reverse Bias barrier കൂട്ടുന്നതിനാൽ വളരെ ചെറിയ reverse current മാത്രം ഉണ്ടാകും.

Carrier movement in PN junction



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4. Quantum Theory, Photon and Photoelectric Effect—

Light energy is exchanged in packets called photons. Photon energy is $E=hf=hc/\lambda$. The Photoelectric Effect is emission of electrons from a suitable surface when incident light frequency exceeds threshold frequency. Intensity controls number of photoelectrons, while frequency controls maximum kinetic energy.

$$hf = \phi + KE_{\max}$$

Malayalam: Suitable metal surface-ൽ Threshold Frequency-നേക്കാൾ കൂടുതലുള്ള frequency-യിലുള്ള light പതിക്കുമ്പോൾ electrons പുറപ്പെടുന്ന phenomenon ആണ് Photoelectric Effect.

Photon-energy calculator

Wavelength λ (nm)

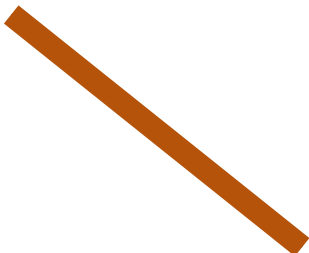
500

Enter wavelength.

Photoelectron emission

Incident photons

Emitted electrons



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5. Laser Principle, Characteristics, Applications and Ruby Laser—

LASER means Light Amplification by Stimulated Emission of Radiation. Laser action requires pumping, metastable state, population inversion, stimulated emission and an optical resonator.

Stimulated absorption: atom absorbs photon and rises to higher level.

Spontaneous emission: excited atom emits randomly.

Stimulated emission: incident photon produces a second identical photon.

Population inversion: more atoms in upper laser level than lower level.

Laser light is monochromatic, coherent, directional, intense and low-divergence. Applications include communication, medicine, cutting, welding, measurement, alignment, barcode reading and data storage.

Ruby Laser: synthetic ruby rod doped with chromium ions is pumped by a flash lamp. One end mirror is fully reflecting and the other partially reflecting. It is a three-level solid-state laser.

Malayalam: Higher energy state-ൽ lower energy state-നേക്കാൾ കൂടുതൽ atoms ഉണ്ടാകുന്ന condition ആണ് Population Inversion. Laser action-ന് ഇത് ആവശ്യമാണ്.

Optical Pumping →

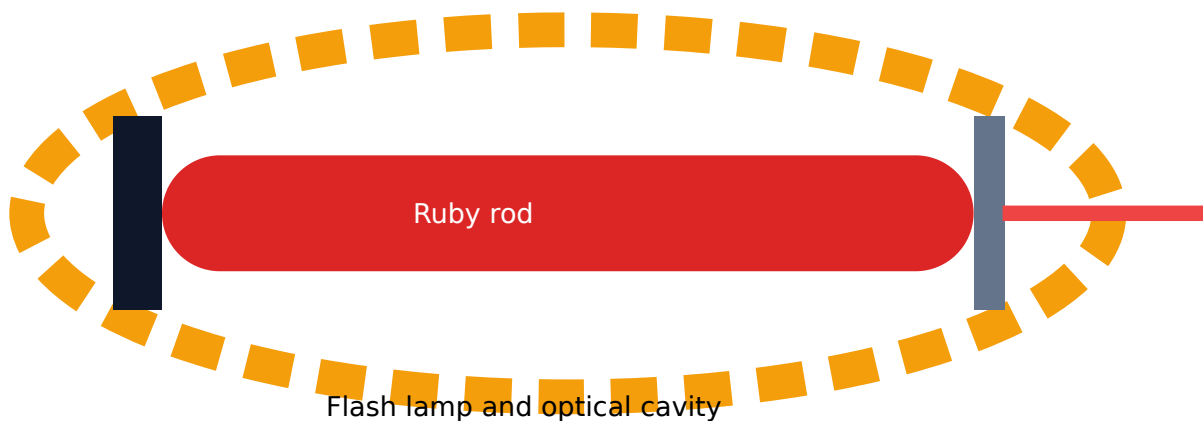
Excited State →

Metastable State →

Population Inversion →

Stimulated Emission →

Laser Output



Construction of a Ruby Laser.

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periments and record format

The official syllabus requires experiments covering all Course Outcomes, with a minimum of 8–10 experiments. Every experiment record should contain Aim, Apparatus, Formula, Theory, Diagram, Procedure, Observation, Calculation, Result, Precautions and Viva.

Safety: Isolate electrical supply before changing diode circuits; handle mercury thermometer and glassware carefully; keep hands clear of flywheel and suspended weights; never look directly into a laser beam.

Module I experiments

Vernier Calipers

Aim: Determine volume of spherical, cylindrical or rectangular body.

Key points: Main Scale Reading, Vernier coincidence, Least Count, Zero Error, corrected dimension and volume.

$$\text{Reading} = \text{MSR} + (\text{VC} \times \text{LC}) \pm \text{zero correction}$$

Screw Gauge

Determine wire diameter, solid-ball diameter or glass-plate thickness. Record pitch, circular-scale reading, least count, zero correction and backlash precaution.

Spherometer

Determine radius of curvature of spherical surface using sagitta and distance between legs.

$$R = \frac{l^2}{(6h)} + \frac{h}{2}$$

Parallelogram Law of Forces

Verify vector addition and find unknown mass. Draw force polygon, record loads and compare experimental resultant.

Module II experiments

Mercury Thermometer

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hot-bath temperature; convert Celsius readings into Fahrenheit and Kelvin.

Hooke's Law

Plot Load versus Extension and determine spring constant from the slope.

$$F = kx$$

Stoke's Law

Measure terminal velocity of a spherical body and determine coefficient of viscosity of the liquid.

U-Tube Relative Density

Measure liquid-column heights and determine relative density using hydrostatic balance.

Module III experiments

Simple Pendulum

Determine g using pendulum length and time period.

$$g = 4\pi^2 l / T^2$$

Moment Bar

Determine unknown mass using equality of clockwise and anticlockwise moments.

Spring Constant by Oscillation

Determine k from mass and oscillation period.

$$k = 4\pi^2 m / T^2$$

Resonance Column

Determine velocity of sound in air at room temperature using tuning fork and resonance length.

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Module IV experiments

V-I Characteristics of Si and Ge Diodes

Wire forward-bias circuit, vary voltage, record current, plot graph and determine knee voltage. Repeat for Ge and Si.

Photoelectric Cell Characteristics

Measure photocurrent for controlled change of light intensity or distance, tabulate readings, plot graph and interpret.

Universal record checklist

- Experiment number and date
- Aim and apparatus
- Formula and theory
- Labelled diagram
- Stepwise procedure
- Observation table with units
- Calculation and graph
- Result and percentage error
- Precautions, troubleshooting and viva
- Faculty signature

EXPECTED QUESTIONS

Practice questions with answer support

These are internally prepared practice questions, not official SITTR questions.

Module I — Mechanics

1 mark Differentiate scalar and vector quantities.

A scalar has magnitude only; a vector has magnitude and direction. Example: speed is scalar, velocity is

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3 marks

State Newton's three laws with one example each.

First law explains inertia; second law gives $F=ma$; third law states equal and opposite action-reaction forces. Add one correct example for each.

7 marks

Explain conservation of momentum and recoil of a gun with equation.

Define momentum, state conservation for isolated system, write total initial momentum = total final momentum, explain bullet-gun opposite momenta and include $m_{\beta}v_{\beta} = -m_{\gamma}v_{\gamma}$.

Numerical

A 5 kg body accelerates at 3 m/s². Find force.

$F=ma=5\times 3=15\text{ N}$.

Diagram

Draw and explain the Parallelogram Law.

Draw two vectors from common origin, complete parallelogram and show diagonal as resultant.

Module II — Matter and Thermodynamics

1 mark

Define coefficient of viscosity.

It is the tangential force per unit area required to maintain unit velocity gradient between adjacent fluid layers.

3 marks

Differentiate streamline and turbulent flow.

Compare particle paths, velocity stability, mixing and energy loss, with examples.

7 marks

State Bernoulli's Theorem and explain the atomizer.

State assumptions and equation, then explain how high air velocity creates low pressure, causing liquid rise and spray.

Numerical

Convert 100 °C into Fahrenheit and Kelvin.

$F=(9/5\times 100)+32=212\text{ °F}$; $K=100+273.15=373.15\text{ K}$.

Application

Why is lubricant viscosity important in machines?

[Skip to handbook](#) nation, friction, wear, heat generation and power loss.

Module III — Rotation and Waves

1 mark **Define radius of gyration.**

It is the distance from the axis at which the whole mass may be assumed concentrated to give the same Moment of Inertia: $I = Mk^2$.

3 marks **State Parallel and Perpendicular Axis Theorems.**

Parallel: $I = I_{cm} + Md^2$. Perpendicular for plane lamina: $I_z = I_x + I_y$.

7 marks **Explain SHM and derive or use expressions for pendulum and spring time periods.**

Define restoring-force condition, mean position, amplitude, time period and frequency; state $T = 2\pi\sqrt{l/g}$ and $T = 2\pi\sqrt{m/k}$ with applications.

Numerical **Find wave velocity for $f = 500$ Hz and $\lambda = 0.68$ m.**

$v = f\lambda = 500 \times 0.68 = 340$ m/s.

Diagram **Draw transverse and longitudinal waves.**

Label crest, trough and wavelength for transverse wave; compression, rarefaction and wavelength for longitudinal wave.

Module IV — Modern Physics

1 mark **What is threshold frequency?**

Minimum incident-light frequency required to cause photoelectric emission from a material.

3 marks **Differentiate intrinsic and extrinsic semiconductors.**

Intrinsic is pure with equal electron and hole concentrations; extrinsic is doped and has majority and minority carriers.

7 marks **Explain construction and working of Ruby Laser.**

Describe ruby rod, chromium ions, flash lamp, two mirrors, three-level pumping, metastable state, population inversion, stimulated emission and output through partially reflecting mirror.

Numerical **Calculate photon energy for wavelength 500 nm.**

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$(6.63 \times 10^{-34} \times 3 \times 10^8) / (500 \times 10^{-9}) = 3.98 \times 10^{-19} \text{ J} \approx 2.48 \text{ eV}$.

Application List four laser applications.

Communication, surgery, cutting/welding, alignment, barcode reading, data storage or measurement.

SELF-LEARNING

Official weekly activity plan

Weeks 1-4 — Module I

1. Classify fundamental and derived quantities; practise Vector Addition and Resolution.
2. Solve Equations of Motion; demonstrate Newton's First Law; collect daily-life Inertia examples.
3. Find Newton's II and III Law examples; list appliance power ratings and calculate one-hour energy use.
4. Calculate Work in climbing stairs or lifting water; relate food energy to daily activity.

Weeks 5-8 — Module II

5. Hooke's Law with rubber band and coins; Elasticity applications; Young's Modulus practice; concept map.
6. Identify Streamline/Turbulent Flow; qualitative continuity; paper-strip Bernoulli activity; applications.
7. Compare viscosity of water, oil and honey; observe drop time; study temperature effect.
8. Ice-water phase-change observation; compare heating of materials; temperature conversions; Heat Transfer examples.

Weeks 9-12 — Official enrichment

Official mismatch note: These activities appear in the official table but are outside the listed core Module III theory.

9. Mirror and light observations.
10. Convex lens, optical instruments, Total Internal Reflection and Optical Fibre.
11. Static Electricity, magnets, appliance current/voltage, Kirchhoff's Laws.
12. Conductor resistivity, Electric Generator and magnetic field due to current.

Weeks 13-16 — Module IV

Skip to handbook or applications, Band Theory, Fermi level enrichment and device study.

14. PN Junction simulation, diode applications, temperature effect, dynamic resistance and saturation

15. Photoelectric demonstration, flow chart, Einstein-equation numericals and quiz.

Course 2002B

16. Spontaneous versus Stimulated Emission, laser applications and Ruby Laser energy-level diagram.

ASSESSMENT

Evaluation and question-paper pattern

Practical Continuous Evaluation — 50

Criterion	Marks
Preparation & Punctuality	10
Experimental Setup & Procedure	10
Observation & Data Recording	5
Analysis & Result Interpretation	10
Viva / Concept Understanding	10
Workmanship, Discipline & Teamwork	5

Practical Test — 40

Category	Marks
Write-up / Procedure	10
Setup & Execution	10
Observation & Result	8
Viva	8
Record Completion	4

End Semester Theory Pattern — 2.5 hours, 60 marks

Module	Part A	Part B	Part C	Total
I	$2 \times 1 = 2$	2 of $3 \times 3 = 6$	1 set with choice = 7	15
II	2	6	7	15
III	2	6	7	15
IV	2	6	7	15
Total	8	24	28	60

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Last-minute checklist

Course 2002B

Module I

- SI units
- Vector laws
- Three motion equations
- Newton's Laws
- Momentum conservation
- Work-Energy-Power

Module II

- Stress, Strain, Moduli
- Continuity equation
- Bernoulli + Atomizer
- Stoke's Law
- Temperature conversions
- Heat-transfer modes

Module III

- Linear-angular relations
- Centripetal Force
- Moment of Inertia axes
- Torque and Angular Momentum
- SHM formulas
- $v=f\lambda$

Module IV

- Energy-band diagrams
- Intrinsic / Extrinsic
- PN Junction biasing
- Diode V-I curve
- Photon and Einstein equations
- Ruby Laser

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Exam discipline: Write the formula first, substitute with SI units, show calculation, box the final answer and label every diagram.

REFERENCES AND EQUIPMENT

Official learning resources

Books

- NCERT Physics for Classes XI and XII, Parts I and II.
- Applied Physics, Volumes I and II, TTTI Publications, Tata McGraw Hill.
- Fundamentals of Physics, Halliday, Resnick and Walker, Wiley India.

Web resources listed in syllabus

- phydemo.app
- phet.colorado.edu

Major equipment for a batch

Vernier Calipers, Screw Gauge, Spherometer, Parallelogram Law apparatus, Mercury Thermometer, Heat Bath, Hooke's Law apparatus, Stop Watch, U-Tube apparatus, Stoke's Law apparatus, Convex Lens set, Prism/Glass Slab, Voltmeter, Ammeter, Rheostat, Battery Eliminator, Standard Resistors, Semiconductor Diodes, Breadboard, Microammeter, Carbon Resistors, Photoelectric Cell apparatus, Simple Pendulum apparatus, Resonance Column, Flywheel, Slotted Weights and Tuning Forks.

Official source declaration: Academic scope follows the supplied SITTTTR Kerala Diploma Curriculum Revision 2026 Course 2002B syllabus. Explanations, diagrams, calculations and practice questions in this handbook are original educational material.